

Nigel Loh

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EDUCATION

University of Toronto Toronto, Canada
Bachelor of Science in Computer Science and Math (Co-op Program) (GPA: 3.96 / 4.0) *Expected June 2028*

- Coursework: Operating Systems, Machine Learning, Deep Learning, Data Structures, Algorithms, Databases, Computer Organization

WORK EXPERIENCE

Capital One | *NestJS, TypeScript, Angular, AWS, Jenkins, PostgreSQL, Claude Code* McLean, Virginia
Software Engineer Intern *June to August 2026*

- Architected a self-service questionnaire platform letting **15+** internal banking teams configure and deploy dynamic intake forms through an Angular interface without engineering support, cutting workflow launch time **95%** (weeks to hours)
- Built REST APIs in NestJS and TypeScript over PostgreSQL for questionnaire schemas and response persistence, modeling reusable sections as UUID references so teams composed forms without duplicating content, sustaining **3M+ API calls** per day
- Designed a **tagging system** replacing the multi-table scan behind combined department and creator filters, resolving questionnaire lookups against a single table; shipped through Jenkins CI/CD to AWS ECS with audit logging
- Wrote Jest unit tests and Cucumber e2e tests, reaching **100%** coverage

PointClickCare | *Python, MySQL, Azure, OpenAI, LangChain* Toronto, Ontario
Machine Learning/Data Engineer Intern *May to August 2025*

- Built a RAG pipeline over **10M+** clinical notes and **5M+** assessment questions using OpenAI text-embedding-3-small, Azure AI Search, and LangChain, cutting a chart review the team reported taking weeks down to minutes
- Validated retrieval quality by manually reviewing a random sample of **3,000+** question matches, tuning the similarity threshold where matches returned too loose
- Built high throughput pipelines with Pandas, NumPy, and MySQL to ingest and preprocess **20M+** clinical records

PROJECTS

Dance Dance Revolution Keyboard (UofTHacks Winner) | *C++, Python, Arduino*

- Led a team of four to build a working dance pad keyboard, writing C++ Arduino firmware that streamed key down and key up events over USB to a Python layer decoding steps as **Braille** chords, with a coalescing window so presses landing milliseconds apart resolved to one character instead of several
- Won **Best Hardware** at UofTHacks13 among 500+ participants by creating an interactive input device that combines movement with productivity

One Million Notes (UofTHacks Winner) | *Next.js, Node.js, Firebase, TypeScript*

- Led a team of three to build **onemillionnotes.co**, a collaborative web app containing one million user-editable notes, winning a prize at UofTHacks12 among 500+ participants
- Implemented **Firebase Firestore** for real-time synchronization, enabling users to instantly create and edit notes with zero reload lag across devices

PokerFish | *React.js, FastAPI, Docker, AWS, PostgreSQL, Redis, PyTorch*

- Implemented real time multiplayer gameplay using **WebSockets** with Redis for state synchronization, a React frontend, and FastAPI backend, persisting game data in PostgreSQL and deployed on AWS EC2
- Trained a poker agent through self-play with **Proximal Policy Optimization** in PyTorch, serving it in game as an adaptive opponent

TECHNICAL SKILLS

- Languages:** Python, TypeScript/JavaScript, Java, C/C++, Assembly, SQL
- Frameworks & Databases:** Node.js, NestJS, React, Angular, FastAPI, PyTorch, PostgreSQL, MySQL, Redis
- Cloud & Tools:** AWS (ECS, EC2), Azure, Docker, Jenkins CI/CD, Git, Linux, Jest, Cucumber, Claude Code